16th IEEE International Conference on Control Applications
Part of IEEE Multi-conference on Systems and Control
Singapore, 1-3 October 2007

Real-Time Experiment of Snore Active Noise Control

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Abstract— **This paper presents real-time experiment of active noise control (ANC) algorithms for reducing nonstationary, intermittent snore. Three snore ANC algorithms have been developed and tested to evaluate their performance. The ANC system uses both finite impulse response (FIR) and infinite impulse response (IIR) adaptive filters. Acoustic feedback was minimized by using both adaptive IIR filter and FIR filter with feedback neutralization. Real-time experiments were conducted using a digital signal processor development system and recorded snore samples. An average reduction of 15-20 dB can be achieved using multi-channel ANC systems.**

I. Introduction

An ANC system generates an anti-noise of equal magnitude and opposite phase and combines it with the primary (unwanted) noise resulting in the cancellation of both noises [1]. Since the characteristic of the noise source and the environment are time varying, most practical ANC systems are adaptive in nature to produce optimum results. Acoustic ANC finds numerous applications in reducing low-frequency noises without the change of installation and configuration of the noise source. One of the very challenging and important applications of ANC is snore reduction. Snore is an annoying low-frequency noise encountered in daily life. With the use of low-cost devices such as the digital signal processors, the implementation of snore ANC systems for cost-effective consumer application is feasible.

The experimental setup of snore ANC system shown in Figure 1 consists of a twin-size bed with wooden headboard. Two speakers (six and half inches each) are mounted on the headboard to serve as secondary speakers for generating anti-noise as shown in Figure 2. Two error microphones are also mounted on the headboard to pick up the residual noise. A KEMAR (Knowles Electronics Mannequin for Acoustics Research) dummy placed on the bed is used as a human model to replicate the bed partner. A primary speaker, which is driven by a digital audio player to play the recorded snore samples, is used to simulate a snorer. A reference microphone placed on the headboard close to the snorer is used to pick up the snore as a reference signal. A floating-point digital signal processor, the TMS320C32, is used for real-time experiments. Two microphones inside the ears of the KEMAR are used to evaluate the performance of snore ANC system perceived by the bed partner.



Fig. 1. Experimental setup of snore ANC system



Fig. 2. Experimental setup of snore ANC system (Front View)

II. Single-channel snore ANC algorithms

This section introduces the adaptive algorithms used for real-time experiments of snore ANC. Both Single-channel and multi-channel feedforward ANC algorithms [1] were used for experiments. The ANC configurations used for experiments include [2], [3]:

1) A single-channel feedforward ANC system with one reference microphone, two secondary loudspeakers connected as one (i.e., the two secondary speakers are driven by the same anti-noise), and two error microphone signals are combined. This system uses the single-channel filtered-X least mean square (FXLMS) algorithm [4].
2) A multi-channel feedforward ANC system, which uses two secondary speakers and two error

Manuscript received May 10th, 2007.

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microphones independently. This multi-channel ANC system uses the 1x2x2 FXLMS algorithm [5].

A single-channel ANC system using the FXLMS algorithm is illustrated Figure 3. In this configuration, the adaptive filter $W(z)$ is used to generate the anti-noise $y(n)$, $d(n)$ is the primary noise to be cancelled, $e(n)$ is the residual noise picked up by the error microphone (In single-channel ANC configuration, the error signal $e(n)$ is the mixed signal sensed by the two error microphones), $\hat{S}(z)$ is the model (or estimate) of the secondary path transfer function $S(z)$, and $P(z)$ is the primary path transfer function.

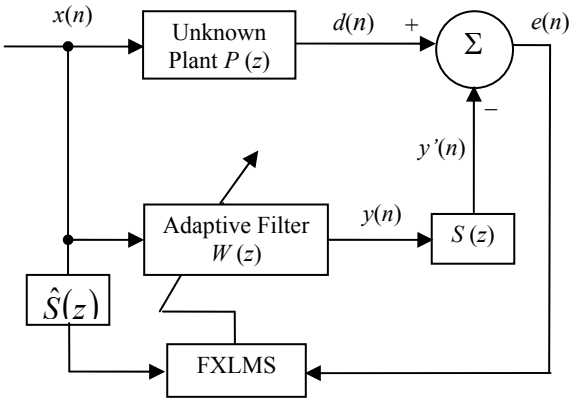


Fig. 3. A single-channel feedforward ANC system with the FXLMS algorithm.

The single-channel FXLMS algorithm is expressed as follows:

$$y(n) = \mathbf{w}^T(n)\mathbf{x}(n) \quad (1)$$

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu e(n)\mathbf{x}(n) * \hat{s}(n) \,, \quad (2)$$

where $\mathbf{w}(n) = [w_0(n)\ w_1(n)\ \ldots\ w_{L-1}(n)]^T$ is the weight vector of the adaptive filter $W(z)$, $\mathbf{x}(n) = [x(n)\ x(n\text{-}1)\ \ldots\ x(n\text{-}L\text{+}1)]^T$ is the signal vector, μ is the step size that determines the stability and convergence rate of the FXLMS algorithm, and $\hat{s}(n)$ is the impulse response of $\hat{S}(z)$. Filter lengths must be long enough to accurately model the paths.

Figure 4 shows the single-channel feedforward ANC system using IIR filter. In this configuration, the adaptive filters $A(z)$ and $B(z)$ are used to generate the anti-noise. The single-channel adaptive IIR algorithm is expressed as follows:

$$y(n) = \mathbf{a}^T(n)\,\mathbf{x}(n) + \mathbf{b}^T(n)\,\mathbf{y}(n-1) \quad (3)$$

$$\mathbf{a}(n+1) = \mathbf{a}(n) + \mu_1 e(n)\mathbf{x}(n) * \hat{s}(n) \quad (4)$$

$$\mathbf{b}(n+1) = \mathbf{b}(n) + \mu_2 e(n)\mathbf{y}(n-1) * \hat{s}(n) \,, \quad (5)$$

where $\mathbf{a}(n)$ and $\mathbf{b}(n)$ are the weight vectors of the adaptive filters $A(z)$ and $B(z)$, respectively. The feedback signal vector is given as $\mathbf{y}(n\text{-}1)=[y(n\text{-}1)\ y(n\text{-}2)\ \ldots\ y(n\text{-}M)]^T$.

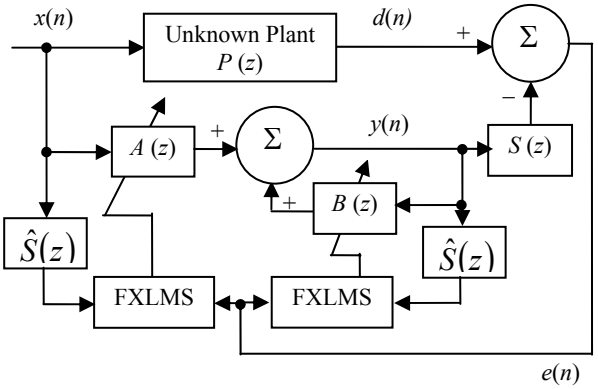


Fig. 4. A single-channel ANC system using adaptive IIR filter.

Figure 5 shows a single-channel ANC system using adaptive FIR filter with feedback neutralization. In this configuration, $\hat{F}(z)$ is an adaptive filter used to model the feedback path, thus canceling the acoustic feedback from the secondary loudspeaker(s) to the reference microphone. The feedback neutralization algorithm is expressed as

$$\mathbf{x}_1(n) = \mathbf{x}(n) - \hat{\mathbf{f}}^T(n)\mathbf{y}(n-1) \,, \quad (6)$$

where $\hat{\mathbf{f}}(n)$ is the weight vector of the feedback path model $\hat{F}(z)$, and $x_1(n)$ is served as the reference signal to the ANC system.

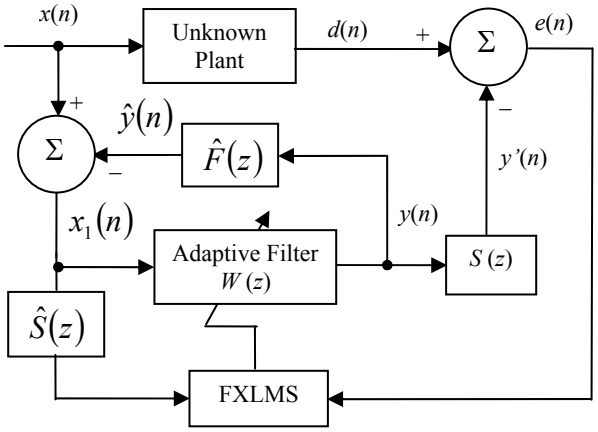


Fig. 5. A single-channel feedforward ANC system with feedback neutralization.

III. Real-Time Experiments of Single-Channel ANC Systems

The ANC algorithm operates in two stages. In the first stage, the secondary path is estimated using an offline modeling technique summarized in [1]. In the offline modeling process, white noise is generated which drives the secondary speakers and also acts as the input signal to the adaptive filter for system identification. The error microphone picks up this excitation signal as the desired signal. The difference between this signal and the output of the adaptive filter is used to update the weight vector of the secondary-path estimation filter using the least mean square

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(LMS) algorithm. The second stage of algorithm is online ANC process summarized in previous section that generates the canceling signal, $y(n)$.

The gains of the power amplifiers driving the secondary speakers are adjusted so that the volumes of the snore [6] and the volumes of the anti-snore played by the secondary speaker are at the similar levels at the error microphone locations. The gains of the preamplifiers for the microphone signals are adjusted to match the peak-to-peak voltage range +/- 3V of the A/D converter on the C32 board for preventing the saturation of the converter. The snore played on the primary speaker is picked up by a reference microphone as input signal $x(n)$ to the ANC system. The ANC algorithm generates the anti-noise $y(n)$ to drive the secondary speakers. The adaptive filter $W(z)$ which generates the canceling signal can be realized using three filters: (1) FIR, (2) IIR, and (3) FIR filters with feedback neutralization. Real-time experiments were conducted for these algorithms. Weights of the adaptive filters are updated only when snore is detected using simple power estimation. After several experiments, a filter length of 256 and step size μ =.0006 was found to give better performance

A. ANC system using FIR filters

The first set of real-time experiment tests the performance of snore ANC system using the single-channel FXLMS algorithms given in equations (1) and (2). When the error microphones were placed near the ears of the KEMAR, the average noise reduction at the right ear of the KEMAR was around 5-10 dB as shown in Figure 6 and at the left ear it was around 2 dB. This is because the primary loudspeaker (snorer) was located at the right side of KEMAR so the error microphone close at the right ear of the dummy picks up the snore signal without any obstacle in the transmission path, whereas the signal picked up by the error microphone placed close at the left ear has the head of the KEMAR as an obstacle. There is a zone of silence around the error microphone called the "quiet zone". As the two error microphones are placed near the ears of the KEMAR, the ears of KEMAR are now in the quiet zone. However, as the two error microphone signals are mixed, the ANC system is dominated by the signal sensed by the right error microphone, thus degrading its performance at the left ear.

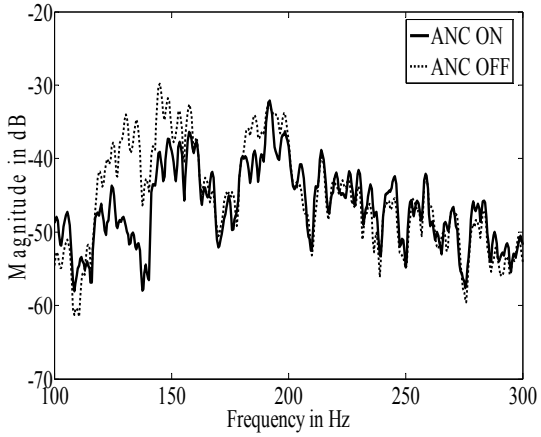


Fig. 6. Performance of snore ANC system at the right ear of the KEMAR using FIR filter.

Figure 6 shows that the single-channel snore ANC system using FIR filter gives better performance in the low-frequency range of 120-160 Hz. When the error microphones were placed on the headboard, the average snore-noise reduction at the right ear of the KEMAR was around 2-5 dB and at the left ear was around 5-8 dB. The achievable noise reduction in this case is decreased because the ears of the KEMAR are far away from the quiet zone centered at the error microphone.

B. ANC system using IIR filters

The adaptive IIR filter with the FXLMS algorithm given in equations (3), (4) and (5) was implemented for real-time experiments. When the error microphones were placed close at the ears of the KEMAR, the average noise reduction at the right ear was around 5-12 dB as shown in Figure 7, which has better performance in the frequency range of 180-200 Hz as compared to Figure 6. There is no significant noise reduction at the left ear of the KEMAR. Even though the error microphones are placed near the ears of the KEMAR, noise reduction is not achieved at the left ear because of the mixing of two signals. When the error microphones were placed on the headboard, the average noise reduction at the right ear of the KEMAR was around 2-5 dB and at left ear was around 5-8 dB.

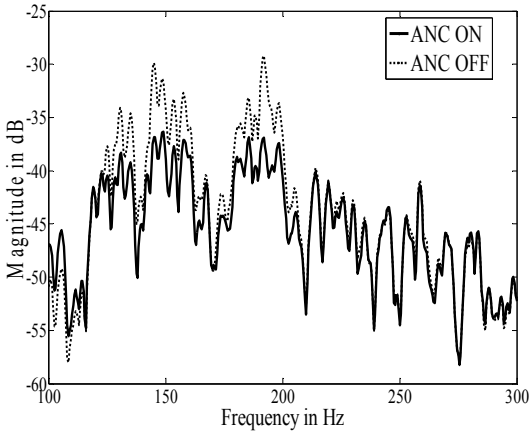


Fig. 7. Performance of snore ANC system at the right ear of the KEMAR using IIR filter.

C. ANC system with feedback neutralization

In additional to equations (1) and (2), the feedback neutralization defined in (6) was implemented for experiments. When the error microphones were placed close at the ears of the KEMAR, the average noise reduction at the right ear of the KEMAR was around 5 dB as shown in Figure 8. Comparing Figures 6, 7, and 8, it is clear that the ANC algorithm with feedback neutralization achieves better performance at high frequencies from 190 to 300 Hz. This is because acoustic feedback is more severe at high frequency.

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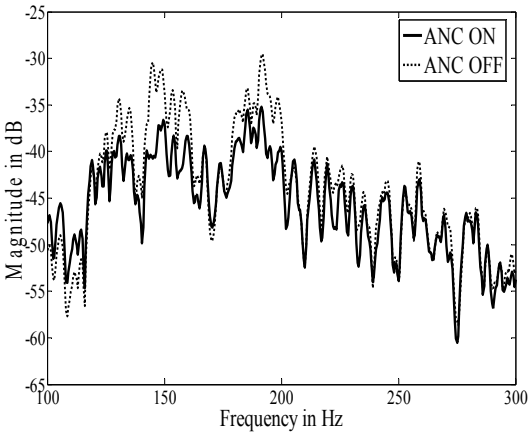


Fig. 8. Performance of snore ANC system at the right ear of the KEMAR using feedback neutralization.

When the error microphones were placed on the headboard, the average noise reduction at the right ear of the KEMAR was around 2-5 dB, and at the left ear was around 5 dB.

D. Comparison of single-channel snore ANC algorithms

Real-time experimental results of single-channel snore ANC systems are summarized in Table I. It shows that there is not much difference in using either adaptive FIR or IIR filters based on overall noise reduction. In the case of FIR filter with feedback neutralization, the overall noise reduction of ANC system was not as good as the IIR and FIR systems, except at the high frequency range. As both FIR and IIR filters are giving the similar results, FIR filters are preferred because of stability reasons.

The disparity of noise reduction at the ears of the KEMAR is because in the single-channel ANC configuration, the residual noises picked up by two error microphones are combined as a single input to the ANC system. This type of system cannot distinguish the left and the right error microphone signals. The right microphone, which is close to the primary speaker, picks up more noise than the left microphone. When the microphone signals are mixed, the signal picked up by the right microphone has dominance over the left microphone signal.

The best noise reduction is obtained near the error microphone location. In the single-channel ANC configuration, even though the error microphones are placed near the ears of the KEMAR, good reduction is not achieved because the error microphone signals are combined. To overcome this drawback, we use the multiple-channel ANC configuration with the 1x2x2 FXLMS algorithm, which uses two error microphones separately and each secondary speaker is driven by independent anti-snore generated by different adaptive filters. In the multiple-channel ANC configuration, the best error microphone positions are near the ears where two individual quiet zones can be formed near the ears. Optimum noise reduction is achieved because the error microphones pick up almost the same signal that is perceived by the ears, and hence more meaningful noise reduction can be achieved.

The parameters that affect experimental results are the filter lengths of the secondary-path estimation filters and the adaptive filters for ANC, the step size for the offline modeling process, and the step size to update the ANC adaptive filters. Different error microphone locations were also tested to optimize the performance of the ANC systems. Of all the parameters, the error microphone location was found to be very critical.

IV. MULTI-CHANNEL SNORE ANC SYSTEM USING 1X2X2 FXLMS ALGORITHM

Figure 9 shows an example of multi-channel feedforward ANC system using the 1x2x2 FXLMS algorithm. It uses one reference microphone, two secondary speakers, and two error microphones. These two error microphones pick up the residual errors $e_1(n)$ and $e_2(n)$ at different positions, thus able to form two individual quiet zones centered at the error microphones. The ANC algorithm used two adaptive filters $W_1(z)$ and $W_2(z)$ to generate two anti-snores $y_1(n)$ and $y_2(n)$ to drive the two independent secondary speakers. In Figure 9, $d_1(n)$ and $d_2(n)$ are the primary noises to be cancelled, $S_{11}(z)$, $S_{12}(z)$, $S_{21}(z)$, and $S_{22}(z)$ are the secondary path transfer functions, and $P_1(z)$ and $P_2(z)$ are the primary path transfer functions.

The 1x2x2 FXLMS algorithm is summarized as follows:

$$y_i(n) = \mathbf{w}_i^T(n)\mathbf{x}(n),\ i = 1, 2 \tag{7}$$

$$\mathbf{w}_1(n+1) = \mathbf{w}_1(n) + \mu_1[e_1(n)\mathbf{x}(n) * \hat{s}_{11}(n) + e_2(n)\mathbf{x}(n) * \hat{s}_{21}(n)] \tag{8}$$

$$\mathbf{w}_2(n+1) = \mathbf{w}_2(n) + \mu_2[e_1(n)\mathbf{x}(n) * \hat{s}_{12}(n) + e_2(n)\mathbf{x}(n) * \hat{s}_{22}(n)], \tag{9}$$

where $\mathbf{w}_1(n)$ and $\mathbf{w}_2(n)$ are weight vectors of the adaptive filters $W_1(z)$ and $W_2(z)$, respectively, μ_1 and μ_2 are the step sizes, $\hat{s}_{11}(n)$, $\hat{s}_{21}(n)$, $\hat{s}_{12}(n)$ and $\hat{s}_{22}(n)$ are the impulse responses of $\hat{S}_{11}(z)$, $\hat{S}_{21}(z)$, $\hat{S}_{12}(z)$, and $\hat{S}_{22}(z)$, respectively.

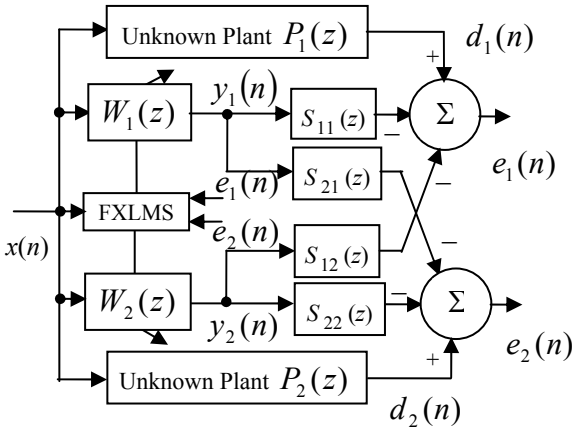


Fig. 9. A multi-channel feedforward ANC system with the 1x2x2 FXLMS algorithm.

The 1x2x2 FXLMS algorithm summarized in equations (7), (8), and (9) was implemented for real-time cancellation of snore. The frequency spectrum shown in Figure 10 was obtained when the error microphones were placed near the ears of the KEMAR. The average noise reduction at both

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ears of the KEMAR was around 15-20 dB. The noise reduction is improved because the algorithm forms two independent quiet zones centered near the KEMAR's ears. This configuration achieves significant noise reduction.

When the error microphones were placed on the headboard, the average snore-noise reduction at the right ear of the KEMAR was around 7-12 dB, and at the left ear was around 2-5 dB. The multi-channel ANC configuration was tested only using FIR filters and the real-time experimental results are summarized in Table II. The average noise reduction for the error microphones when placed near the head of the KEMAR is around 18-20 dB.

From Table II, it is clear that the optimum performance has been achieved when the error microphones are placed near the KEMAR's ears using the 1x2x2 configuration. The average difference in noise reduction between placing the error microphone near the KEMAR's ears and on the headboard is 10-15 dB for the left ear, and 5-8 dB for the right ear of the KEMAR.

Practically, the error microphones cannot be placed near the ears because it requires the snorer's bed partner to wear wireless microphones that are near the ears.

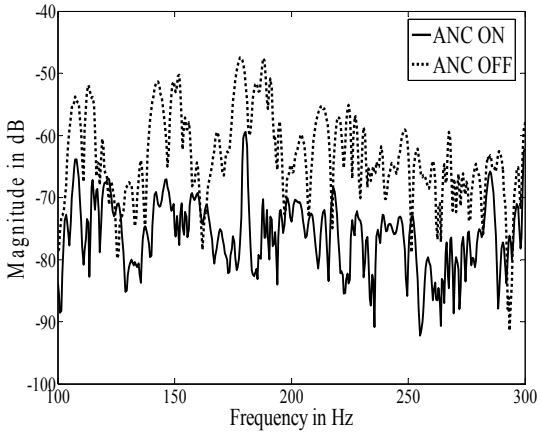


Fig. 10. Performance of snore ANC system at the right ear of the KEMAR using 1x2x2 FXLMS algorithm

TABLE II
PERFORMANCE COMPARISON OF MULTI-CHANNEL ANC SYSTEM USING 1X2X2 FXLMS ALGORITHM

Error Microphone Position	Microphone at KEMAR Ear	Noise Reduction in dB
Near KEMAR Ears	LEFT	15-20
	RIGHT	15-20
On the Headboard	LEFT	2-5
	RIGHT	7-12

V. CONCLUSIONS

This paper presented real-time experiments of snore ANC systems. Three single-channel ANC algorithms were tested in real time and spectral plots were used for comparing the experiment results. The multi-channel ANC system with 1x2x2 configuration was found to be the best. An average noise reduction of about 15-20 dB was achieved. Experimental results showed that the performance of snore ANC system can be improved by using multi-channel algorithms and placing error sensors near the desired quiet zones.

REFERENCES

[1] S. M. Kuo and D. R. Morgan, *Active Noise Control Systems Algorithms and DSP Implementations*, New York, NY: Wiley, 1996.
[2] S. R. Chakravarthy, *Active Snore Noise Control Systems,* M.S. Thesis, Northern Illinois University, Dec. 2005.
[3] S. M. Kuo and S. R. Chakravarthy, "Application of Active Noise Control for Reducing Snore", in *Proc. of ICASSP*, vol.5, pp. 305-308, 2006.
[4] D. R. Morgan, "An Analysis of Multiple Correlation Cancellation Loops with a Filter in the Auxiliary Path," *IEEE Trans*. on *Acoust. Speech Signal Processing*, vol. ASSP-28, pp. 454-467, Aug. 1980.
[5] S. M. Kuo and D. R. Morgan, "Active Noise Control: A Tutorial Review"*, Proc. of IEEE*, vol. 8, pp. 943-973, June 1999.
[6] Digitized wave files by Computer Audio Engineering, available from website: www.caeaudio.com.

TABLE I
PERFORMANCE COMPARISON OF SINGLE-CHANNEL ANC SYSTEMS

Error Microphone Position	Microphone at KEMAR Ear	Noise Reduction in dB for FIR Filter	Noise Reduction in dB for IIR Filter	Noise Reduction in dB for FIR Filter with Feedback Neutralization
Near KEMAR Ears	LEFT	2	0	0
	RIGHT	5-10	5-12	5
On the Headboard	LEFT	5-8	5-8	5
	RIGHT	2-5	2-5	2-5

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